Dark Pattern Radar: Audit & Model Card

E-commerce dark pattern text classifier · TF-IDF + Logistic Regression

Model details

Task	Binary text classification: dark pattern (1) vs not (0) on shopping-site text segments
Model	TF-IDF (1-2 grams, min_df=2, sublinear tf) + Logistic Regression (C=3). Code: pipeline.py
Data	Real dataset, yamanalab/ec-darkpattern: 2,356 unique segments, balanced classes. Duplicates dropped.
Split	80/20 grouped by page_id (seed 42): train 1,883 / test 473. No site in both sets.
Not used	Synthetic xlsx (only 94 unique texts, labels unvalidated).

Overall performance (test set)

Accuracy	Dark precision	Dark recall	Dark F1
0.941	0.948	0.932	0.940

Audit 1: by pattern category (dark examples, share caught)

Category	n	Accuracy
Scarcity	81	100.0%
Social Proof	67	97.0%
Misdirection	42	78.6%
Urgency	34	94.1%
Obstruction	8	87.5%
Sneaking	2	50.0%
Forced Action	1	0.0%

Audit 2: by text style (length)

Style	n	Accuracy	F1
Long (>5 words)	222	0.923	0.943
Short (<=5 words)	251	0.956	0.935

Finding: strong on keyword-driven patterns (Scarcity, Social Proof, Urgency) but weak on Misdirection, where wording is subtle. Overall accuracy hides this gap. Categories with fewer than 10 test examples (Obstruction, Sneaking, Forced Action) are too small to trust.

Explainability: top logistic-regression features

Dark: left, in, only, purchased, bought, no, limited, people, hurry, stock, no thanks, minutes.
Not dark: and, remotes, all, shop, of, new, view, cookies, accessories, new in, share, stay.
The model learned real cues (stock counters, 'X people bought', countdown wording) plus product-navigation words as 'not dark'.

Sample errors

Text	True	Category
I'll Shop Without A Discount	Dark	Misdirection
Long Dress Style Beach Cover Up With Lace Detail Jessica in Metairie, United States	Dark	Social Proof
Add and save!	Dark	Misdirection
Our most recent customers are from New Rochelle, Roseville, and Brooksville	Dark	Social Proof

Limitations

1. **Keyword reliance:** the model matches surface cues, so subtle or reworded patterns (e.g. 'I'll Shop Without A Discount') are missed, and innocent text with cue words can be flagged ('Order by these dates to get it by Valentine's Day').
2. Rare categories have under 30 training examples.
3. Text only: no layout, colour or UI context, which many dark patterns rely on. English, one-time scrape; may not transfer to other sites.